

THE CASE OF THE MISSING PODS

Step-by-step cheat sheet for facilitators. Defaults assume release name `case` and namespace `case`. Swap in your own passphrases if you ran `--set`.

0. Install & verify

From the project root - the directory that contains `missing-pods/`.

```
helm lint missing-pods
helm install case ./missing-pods --namespace case --create-namespace

# Wait for everything to reach Running / Completed (~60 s)
kubectl get pods -n case

# Optional: ping Precinct HQ via the helm test hook
helm test case -n case
```

1. Briefing

~3 min

TEACHES pods, logs

```
kubectl get pods -n case
kubectl logs case-file -n case
```

The case file prints the story and tells you to find the witness.

2. Level 1 - The First Witness

~4 min

TEACHES label selectors, `--show-labels`, `describe`, annotations

```
kubectl get pods -n case -l role=witness --show-labels
kubectl describe pod bystander-alpha -n case
```

Clue is in the `case.clue` annotation. Passphrase: **BADGE-7749**

```
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=BADGE-7749'
```

3. Level 2 - Surveillance Feed

~5 min

TEACHES multi-container logs, `-c`, `--tail`, `grep`

```
kubecttl logs surveillance-cam -n case # noisy, no clue
kubecttl logs surveillance-cam -n case -c noise # static
kubecttl logs surveillance-cam -n case -c camera --tail=20 # the clue
```

Passphrase: **CAMERA-NOIR**

```
kubecttl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=CAMERA-NOIR'
```

4. Level 3 - The Filing Cabinet

~5 min

TEACHES ConfigMaps, `-o yaml`

```
kubecttl get configmap filing-cabinet -n case -o yaml
```

Three files say `DEAD END`. The real clue is in `file-B.txt`.

Passphrase: **FILING-CABINET-B**

```
kubecttl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=FILING-CABINET-B'
```

5. Level 4 - Cracking the Safe

~6 min

TEACHES Secrets, `-o jsonpath`, base64 decoding

```
kubectl get secret evidence-locker -n case \
  -o jsonpath='{.data.note}' | base64 --decode ; echo
```

Passphrase: **OPEN-SESAME**

```
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=OPEN-SESAME'
```

6. Level 5 - The Hidden Suspect

~6 min

TEACHES Deployments, ReplicaSets, `rollout history`, `scale`

```
kubectl get deployments -n case # informant: 0/0
kubectl rollout history deployment/informant -n case # suspicious change-cause
kubectl scale deployment/informant -n case --replicas=1
kubectl wait --for=condition=Ready pod -l role=informant -n case --timeout=60s
kubectl logs -l role=informant -n case --tail=200
```

Passphrase: **INFORMANT-TALKS**

```
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=INFORMANT-TALKS'
```

7. Level 6 - The Getaway Route

~6 min

TEACHES Services, ClusterIP, `exec`, in-cluster DNS, `curl`

```
kubectl get svc -n case
kubectl describe service getaway-svc -n case
kubectl get endpoints getaway-svc -n case
kubectl exec -it detective-terminal -n case -- \
  curl -s http://getaway-svc.case.svc.cluster.local
```

Passphrase: **GETAWAY-CAR**

```
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=GETAWAY-CAR'
```

8. Level 7 - The Vault

~5 min

TEACHES volumes, volumeMounts, mountPath, exec into the filesystem

```
kubectl describe pod vault-keeper -n case | sed -n '/Volumes/,/Conditions/p'
kubectl exec vault-keeper -n case -- ls -la /evidence
kubectl exec vault-keeper -n case -- cat /evidence/manifest.txt
```

Passphrase: **VAULT-OPENED**

```
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/submit?clue=VAULT-OPENED'
```

9. Finale - The Accusation

~6 min

TEACHES helm list, helm get manifest, helm get values

```
helm list -n case
helm get values case -n case          # spot the worker on the wrong team
helm get manifest case -n case | grep -E "(name:|team:)" | head -40
```

Four workers. Three are team: platform. **ledger-worker** is team: contractor - that's the impostor.

```
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/accuse?name=ledger-worker'
```

Victory banner + elapsed time prints. Case closed.

Hints & status (any time)

```
# Status board + ASCII progress bar
kubectl exec -it detective-terminal -n case -- \
  curl -s http://precinct-hq.case.svc.cluster.local:8080/

# Graduated hint for a level (1..7) - 1 = mild, 3 = near spoiler
kubectl exec -it detective-terminal -n case -- \
  curl -s 'http://precinct-hq.case.svc.cluster.local:8080/hint?level=3'
```

Reset / uninstall

```
helm uninstall case -n case
kubectl delete namespace case      # optional, only if you created it

# Fresh start - clean reset, the clock starts over
helm install case ./missing-pods --namespace case --create-namespace
```

Reskin / randomize for a fresh cohort

```
helm install case ./missing-pods --namespace case --create-namespace \
  --set answers.level1=ZEBRA-1 \
  --set answers.level2=ZEBRA-2 \
  --set answers.level3=ZEBRA-3 \
  --set answers.level4=ZEBRA-4 \
  --set answers.level5=ZEBRA-5 \
  --set answers.level6=ZEBRA-6 \
  --set answers.level7=ZEBRA-7 \
  --set answers.culprit=auth-worker
```

Passphrase quick-reference (defaults)

LEVEL	PASSPHRASE	SOURCE
1	BADGE-7749	witness annotation
2	CAMERA-NOIR	surveillance-cam logs (-c camera)
3	FILING-CABINET-B	filing-cabinet ConfigMap, file-B.txt
4	OPEN-SESAME	evidence-locker Secret (base64)
5	INFORMANT-TALKS	informant pod logs (after scale)
6	GETAWAY-CAR	getaway-svc HTTP response
7	VAULT-OPENED	/evidence/manifest.txt
Culprit	ledger-worker	only worker with team: contractor

The Case of the Missing Pods - Helm chart - cheat sheet generated alongside the chart. Defaults shown here apply when no `--set` overrides are used at install time.